Chapter 2

Turbulence and Planning

2.1 Introduction

In Chapter 1, an argument was made that many environmental and resource management situations are characterized by complexity and uncertainty. Indeed, many environmental and resource problems have been characterized as "wicked", "messes" or "metaproblems", to reflect the high degree of complexity involved. Furthermore, the presence of risk, uncertainty and ignorance ensures that solutions will usually be developed on the basis of incomplete knowledge and understanding of natural and social systems. And yet, life cannot stop until complete understanding is available. If decisions were only to be made after complete understanding were achieved, few decisions would ever be taken. We must be willing to take decisions, accepting that we will make errors. However, acknowledging the presence of complexity and uncertainty may allow us to approach decision making somewhat differently than in the past. It may also lead us to focus and structure our planning and decision-making organizations differently.

In this chapter, consideration is given to the importance of *turbulence* with regard to complexity and uncertainty. Consideration also is given to the concept of *chaos*. Attention is then given to the idea of the *precautionary principle*, as well as the concepts of *hedging* and *flexing*. Then, *alternative approaches to planning* are examined, with particular regard to their assumptions about the best way to address complexity and uncertainty.

2.2 Turbulence

Turbulent conditions seem to be more the norm than the exception. For example, at the beginning of the 1990s, who would have predicted that the

Box 2.1 Changing conditions

To be in the midst of change yet oblivious to it is characteristic of the human condition.

Source: Kates, 1995: 627.

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Box 2.2 Future shock

. . . I coined the term "future shock" to describe the shattering stress and disorientation that we induce in individuals by subjecting them to too much change in too short a time.

Source: Toffler, 1971: 3.

former USSR would break up as a country? Who would have forecast that Nelson Mandela would become the leader of South Africa? While in hindsight such events can be understood, ahead of time or while they are unfolding, they emerge as surprises for many people. They were not predicted much in advance. As a result, turbulence can create bewilderment, anxiety and even suspicion.

Turbulence occurs due to complexity and uncertainty, and due to rapidly changing conditions and associated conflicting interests and positions. In this context, Trist (1980) suggested that, over time, four different kinds of planning or decision-making situations can be identified. Trist's analysis reminds us of the necessity to understand the circumstances or contexts which surround planning and management initiatives.

Type 1: Type 1 is the simplest decision-making environment. It is one in which opportunities and problems are randomly distributed and organization is simple. The best strategy involves each person doing the best as possible on a local scale. A market analogue would be a situation of "perfect competition", represented by a world of small factories, family farms and corner stores. The planning style is *inactive*, or one in which the present is judged to be better than the past or the future. The conventional wisdom is captured by expressions such as it is better to wait and see, look before you leap, or let sleeping dogs lie. This is primarily a non-planning world.

Type 2: In the second decision-making environment, conditions continue to be relatively stable or unchanging. However, opportunities and problems start to group or cluster. The best strategy is to find the optimal location to take advantage of opportunities. While the market analogue for Type 1 was "perfect competition", for Type 2 it is "imperfect competition" with businesses becoming more specialized as they seek a comparative advantage. Here, the planning style is *reactive*, with the past regarded as preferable to the present or the future. In this environment, folk wisdom stresses the good old days, or paradise lost. Planning is used here, as there is a best place to find.

Type 3: At this stage, one which Trist called "disturbed reactive conditions", the environment is dynamic and changing. Other firms or competitors seek and attempt to get the same optimal location. Firms are continually reacting to initiatives of other firms. The best strategy is to amass power, as competitive challenges can best be met through accumulated resources and abundant expertise under the firm's control. Bigger is better. The best market analogy here is "oligopoly" in which there are relatively few but large and specialized firms.

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This is the world of large industrial conglomerates or consortia, and of equally large government departments, in which emphasis is placed upon centralized control. In this situation, the future is viewed as better than either the past or the present. Consequently, emphasis is on improving the capacity to predict and to prepare. The art of the calculable becomes valued, with emphasis on econometric models, technical forecasts, operations research and simulations. As in the second type, planning is important. Attention is given to judging where the best waves are going to come from, and then to ensure that you ride the best one and keep others off it. The emphasis is upon technocratic planning.

Type 4: Trist called the fourth type a "turbulent" environment or field. In this type, many large and competing organizations act independently and take actions in diverse directions. The outcome is unanticipated and discordant consequences shared by all. As Trist (1980: 117) remarked, "the result is contextual commotion – as if 'the ground' were moving as well as the organizations. This is what is meant by turbulence". No organization is so big that it can control things, nor is able to do things on its own. A new value becomes collaboration, reflected through consortia-building and sharing, all of which conflict with the basic values found in Type 3 approaches. For Type 4 there is no ready market analogy, other than the concept of *macroregulation*, or increasing involvement by the government in a collaborative manner with the private sector to try and achieve greater stability.

The planning style therefore is best described as *interactive* (Box 2.3). Neither the past, present or future are particularly attractive. Any desirable future will depend on people making it occur, thereby emphasizing the need to make choices among alternative futures. As a result, the interest is less upon forecasting most likely futures, but in creating most desirable futures (a distinction discussed more in Chapters 3 and 4). However, creating a desirable future cannot be done either alone, in isolation from, or in competition with, others. This is why an interactive and collaborative approach to planning and management is so necessary.

Having identified the four types of planning or management situations, Trist then examined the implications, which are several. First, the Type 4 situation requires a different approach to planning and problem solving. Trial-and-error approaches work well in Type 1 conditions. What might be called "craft knowledge" is appropriate in Type 2 situations; the scientific method has most usually been applied in Type 3 circumstances. But the complex

Box 2.3 Interactive planning

It [interactive planning] requires the collaboration of interest groups . . . ; the identification of shared values . . . ; continuous learning . . . ; and continuous evaluation and modification. . . . It is an open-ended unfolding process. [It has been called] adaptive planning.

Source: Trist, 1980: 119.

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Box 2.4 Institutional gaps

The objective of sustainable development and the integrated nature of the global environment/development challenges raise problems for institutions, national and international, that were established on the basis of narrow preoccupations and compartmentalized concerns. Governments' general response to the speed and scale of global changes has been a reluctance to recognize sufficiently the need to change themselves. The challenges are both interdependent and integrated, requiring comprehensive approaches and popular participation.

Yet most of the institutions facing those challenges tend to be independent, fragmented, working to relatively narrow mandates with closed decision processes. Those responsible for managing natural resources and protecting the environment are institutionally separated from those responsible for managing the economy. The real world of interlocked economic and ecological systems will not change; the policies and institutions concerned must.

Source: World Commission on Environment and Development, 1987: 9.

interdependencies of Type 4 situations are usually not amenable to a scientific approach, when that is interpreted as breaking a problem into its component parts. The complex "metaproblems" require a systems approach in which attention is given to relating parts to the whole, and organizations to their larger environments. In Trist's (1980: 119) words, the systems approach is "synthetic, holistic, seeking to capture the Gestalt of system connectedness".

A second implication relates to the nature of public agencies, or the "bureaucracy". The dominant form of bureaucracy in most countries is one which is technocratic, singular (pursues only it own objectives), specialized and centralized (see the comments from the Brundtland Commission in Box 2.4). This approach is best suited to deal with Type 3 conditions. However, as Trist observed, most environmental and resource problems involve Type 4 conditions. The outcome is that we too often rely upon a bureaucracy designed for conditions that no longer exist. And, as we know from military history, being prepared for conditions which no longer exist can be disastrous. For example, after World War I the French built the Maginot Line, an elaborate defensive system of trenches and bunkers, to protect themselves from the Germans. The Maginot Line was based on fighting styles in World War I, when armies occupied trenches opposite from each other, and ground troops skirmished and attacked from the trenches. However, early in World War II, during 1939 and 1940 the Germans used the mobility of their Panzer tanks to sweep through Belgium and into France, and simply outflanked the Maginot Line, making it an anachronism. Resource and environmental management should not be like the Maginot Line – we should be preparing for conditions in the future, not those from the past which may no longer be as relevant.

A third consequence of the Type 4 situation is that the style of comprehensive planning which involves creation of a master plan or blueprint, and which often is very useful in Type 3 situations, will be less appropriate, if appropriate

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Guest Statement

Change, turbulence and uncertainty in Nigeria

Peter Adeniyi, Nigeria

Nigeria, located between latitude 4° N to 14° N and longitude 3° E to 15° E, covers a total area of about 923,769 km^2. The country is characterized by diverse and complex biophysical resources, ethnic groups numbering over 400, and varying socio-economic regimes. Its current estimated population is about 110 million, with urban areas accounting for about 42%. The density of population ranges from 25 to over 2,000 person/km^2.

Nigeria is generally considered to be endowed with abundant human and material resources. At independence in 1960, agriculture, which accounted for more than 50% of gross domestic product (GDP) and over 75% of export earnings, was the most important sector of the Nigerian economy. The discovery of crude oil in the late 1960s and its exploitation in the early 1970s changed the situation, and Nigeria moved from a position of food self-sufficiency to one of heavy dependence on importation of food and manufactured goods. The importation of food items, for instance, increased from US$353.9 million in 1995 to US$1.6 billion in 1998. Yet only 34 million hectares of the 71.2 million hectares of cultivable land are under cultivation. The cost of imported manufactured goods increased from US$529.4 million in 1995 to US$2.2 billion in 1998.

Inspite of the huge income derived from petroleum resources, which now accounts for over 90% of the total annual national revenue, Nigeria's development has largely remained stunted. Most of the over 100,000 communities in the country lacked basic physical and social infrastructures. In 1992, only about 49% of the communities were served with potable water, while 31, 12, 8, 4 and 2% of the communities were served with educational, health, electricity, communications and banking facilities, respectively. This situation is further compounded by brain-drain (i.e. the migration of many capable Nigerians to other countries) and the unprecedented depletion, destruction and degradation of the environmental resource base, leading to increased scarcity; and by unequal access to renewable resources by the rural majority, increasing poverty and frequent violent ethnic conflicts.

The major reasons for this state of affairs include:

(1) Lack of vision and national consensus regarding the role of resource and environmental management.
(2) Political instability. Within 40 years of independence, government has been changed eleven times, eight by the military which governed for 28 years.
(3) Frequent administrative changes. At independence, Nigeria had three regions which were increased to four in 1963. The four regions were divided into 12 states in 1967, 19 in 1976, 21 in 1978, 30 in 1987 and 36 in 1996. The 330 Local Government Areas in 1978 have now become 774. This proliferation led to poor and weak institutions, premature amalgamation of institutions and scrapping of programmes, as well as the instability of programmes and personnel.

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(4) Uncritical adoption of external development policy models and programmes which are often uniformly and mechanically implemented mostly by public officials without due consideration of local environmental conditions, local knowledge systems, and the loose structure and capacity of agencies.
(5) Overbearing power of the Federal Government over resources, lack of power at the lower tiers of government, and inadequate involvement of the private sector, non-governmental organizations (NGOs), community leaders and women.
(6) Poor institutional mandate, lack of coordination and cooperation among public organizations at all levels, and lack of respect for the local people by public officials.
(7) Lack of relevant and up-to-date data on all components of natural resources and the environment and the persistent accumulation of disparate data which are inconsistently named, improperly documented; highly fragmented in different locations, very low in integrity; not organized, and cannot be electronically processed or distributed; and generally are unreliable, incomplete and very difficult to identify, access, use and maintain.
(8) Lack of accountability, transparency, proper evaluation of programmes' outcomes and opportunity to learn from past mistakes.

The present civilian administration in Nigeria seems to be aware of the above pitfalls and the national sustainability imperatives of finding lasting solutions to them. The concepts, principles and practices of resource and environmental management, noted in this book, provide a solid foundation for the resolution of these challenging problems. Above all, however, is the need to recognize the significance of the problems, to mount a sustained commitment to addressing them, and to take a long-term perspective by not only the academic community, but also by all those who formulate development policies and those who take development decisions in Nigeria.

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at all. As Trist argued, the level of change created by an ever-increasing rate of change is too great, and plans too often are outdated before they can be implemented (see Chapter 12 which discusses implementation). Planning for Type 4 conditions requires continuous adaptive planning: *continuous* because frequent modifications will be essential; *participatory* because all stakeholders must have a role (see Chapter 8 about partnerships); *integrated* because various interests must be incorporated; and *coordinated* because the interdependence of issues and decisions must be recognized. A key lesson is that during planning there must

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Box 2.5 Chaos

. . . chaos is not anarchy or randomness. Chaos is order, but it is order that is "invisible." Nor is chaos merely the result of "noise", or interference, or even insufficient knowledge. What chaos implies is a kind of inherent "uncertainty principle" – not just in how we perceive the world but in how the world actually works.

Source: Cartwright, 1991: 44–5.

be equal attention to the *process* as well as to the *product*. Another lesson is that ends need to be agreed upon before means are examined: this is one of the points highlighted in both Chapters 3 and 4. Many of the issues noted above are highlighted in the guest statement by Peter Adeniyi, as he reflects on the experience with resource and environmental management in Nigeria.

2.3 Chaos

If turbulence contributes to complexity and uncertainty, the concept of chaos is even more unsettling. Chaos is order without predictability. In other words, some physical and social systems might be capable of being understood, in the sense that they can be described relative to a set of conditions or rules, but they remain fundamentally unpredictable.

Cartwright (1991) concluded that many of the problems that planners must handle – population growth, land use patterns – may well reflect chaos. In his view, while we may understand the "rules" that influence behaviour at the individual or "local" level, we usually cannot predict the outcomes or impacts at a global scale. The implications of such a conclusion are profound. As Cartwright (1991: 45) remarked, "No matter how much data we gather, no matter how global and complete our models, no matter how rigorously we test them, even so, according to chaos theory, prediction may in some cases be beyond our grasp".

As will be seen later in Section 2.6 on planning models, planners have developed approaches that recognize situations in which there is incomplete understanding. However, these approaches have emerged for pragmatic reasons: there is not always sufficient time to get all the facts; it is often too costly to obtain all the facts; we lack the skill or means to obtain all the facts. In contrast, chaos theory indicates that trying to predict the future is not just impractical in some cases – rather, it is logically impossible.

More attention will be given to predictions or forecasting in Chapter 3. Nevertheless, three conclusions can be reached if it is accepted that chaotic systems are inherently unpredictable and are impossible to understand fully (Cartwright, 1991: 53–4). First, collecting more information and creating more sophisticated models for chaotic systems will likely have little value. Indeed, "research" might be unhelpful if it creates an unrealistic belief about

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Box 2.6

While the prediction of chaotic behavior may be impossible, understanding the order that gives rise to it may not be as difficult as we thought. Highly complex and unpredictable behavior, in other words, can be the product of quite simple and accessible rules.

Source: Cartwright, 1991: 45.

the capability of planning. Planning strategies that assume the capacity for foresight will be inappropriate and can even be misleading at times.

Alternatively, planners must begin to work with not one or two forecasts about the future, but rather should use an ensemble or suite of forecasts. Planners have normally tested models or forecasts by using different assumptions about future conditions and considering the consequences for the results. It is much less common for planners to begin, or run, their alternative models from different "starting points". It is normally assumed that we know what the present conditions are, and we do not consider alternative perspectives about them. Chaos theory challenges the notion that present conditions are known, and also the idea that even if different starting points were chosen there were would unlikely be major differences in outcomes. As a result, it becomes very important to be willing and able to explore which initial starting points have similar trajectories and which do not. Rather than investing more time and effort in collecting more detailed data, or fine tuning our models, we should look for patterns of behaviour, or points to which systems tend to return, even if we cannot readily predict them.

Second, the concept of chaos requires re-examination of some of our ideas regarding the virtues of order and predictability, and the messiness of chaos and disorder. It is possible that a chaotic environment could be as, if not more, desirable than a neat or orderly one.

Third, we may need to accept that chaotic systems are only predictable at a local scale or on an incremental basis. At a global scale or comprehensive basis, chaotic systems are not predictable due to the cumulative effects of many different kinds of feedback within them. However, at a local or incremental level, the consequences of feedback over time may be identifiable. This conclusion becomes a strong argument for using planning strategies or models which are incremental rather than comprehensive, and based on a capacity to be adaptive rather than structured and "blue printable". Thus, as the comments in Box 2.6 indicate, there are grounds to be optimistic and confident regarding handling chaotic systems.

2.4 The precautionary principle

In Germany, starting in the 1950s, there was discussion focused on the need for *Vorsorge*, or foresight. Of particular interest was recognition of the need for

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Box 2.7 Statements of the precautionary principle

Bergen Declaration (ECE), 16 May 1990

In order to achieve sustainable development, policies must be based on the Precautionary Principle. Environmental measures must anticipate, prevent and attack the causes of environmental degradation. Where there are threats of serious or irreversible damage, lack of full scientific certainty should not be used as a reason for postponing measures to prevent environmental degradation.

"Maastricht Treaty", Treaty on European Union, February 1992, Article 130r, paragraph 2

Community policy on the environment shall . . . be based on the Precautionary Principle and on the principles that precautionary action should be taken, that environmental damage should as a priority be rectified at source and that the polluter should pay.

Rio Declaration, June 1992, Principle 15

In order to protect the environment, the precautionary approach shall be widely applied by States according to their capabilities. Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation.

caution, and to move away from always being reactive in planning and management. Various German governments formally incorporated the need for foresight into their procedures, and this later became known as the *precautionary principle*. This approach soon began to be included in international agreements, especially those regarding marine pollution (Box 2.7). In 1992, Principle Fifteen of the Rio Declaration on Environment and Development from the Earth Summit endorsed the precautionary principle.

The precautionary principle reflects the adage that an ounce of prevention is worth a pound of cure. It also stipulates that rather than waiting for complete understanding, or certainty, managers and decision makers should anticipate potential harmful environmental impacts from actions, and take decisions to avoid such harm. It also recognizes that uncertainty is reality, due to our lack of complete knowledge about ecosystem behaviour; faulty assumptions about ecosystem functions; inability to predict the size, needs and desires of future populations; and difficulty in forecasting future technical innovations. Given such uncertainties, the precautionary principle should be considered when making resource decisions in which (1) the range of possible impacts from one or more uses cannot be predicted, (2) one or more of the outputs or outcomes could have extremely undesirable impacts for future people, and (3) substitutes are not available for the resource to be used.

Young (1993: 15–16) suggested three possible interpretations related to the precautionary principle. These are:

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- a conservative interpretation, in which use or activity is approved only if it poses no danger to an ecological system or does not reduce environmental quality, and is confined within boundaries that permit complete reversibility;
- a more liberal interpretation, in which uses or industries judged to be "risky" are required to use the best available technology, and in addition a precautionary safety margin is established which keeps ambient environmental concentrations well below a specified acceptable threshold; and
- a relatively weak interpretation, in which the requirement is the use of best available technology that does not involve undue expense.

These options illustrate that there is as yet no consensus regarding a definition of the precautionary principle. However, this situation is not necessarily bad, as it does provide scope to custom design its application relative to local needs, conditions and circumstances. Young (1993: 14, 16) provided sensible advice regarding such questions. In his view, the principle should be used with actions involving adverse outcomes, particularly when such outcomes are suspected to be of a catastrophic kind. In this manner, its use should force debate about the types and magnitudes of human-induced change to the environment judged to be acceptable by a society. In addition, he commented that application of the principle should never mean that all developments with uncertain environmental impacts should be stopped. What it does mean, however, is that when a possibly irreversible action that could have unpredictable consequences in the future is being contemplated, all alternative options need to be considered before taking a final decision. He also concluded that such irreversible actions should only be undertaken when it is concluded that not doing so could impose significant social costs on the present generation.

Nevertheless, as Bodansky (1991: 43) cautioned, use of the precautionary principle does not provide a guarantee that serious environmental harm will never occur (Box 2.8). Many contemporary environmental problems were not anticipated. In addition, even if regulators had opted for a cautious approach, it is likely that many of the problems would not have been prevented. For example, chlorofluorocarbons (CFCs) and dichlorodiphenyltrichloroethane (DDT) were judged to be environmentally benign when they were first developed. Problems emerged later not because the regulators had approved their use in the context of uncertainties, but because scientists had not tested for the types of environmental impacts that later became a problem. Thus, extensive tests had been

Box 2.8 Concerns

Although the precautionary principle provides a general approach to environmental issues, it is too vague to serve as a regulatory standard because it does not specify how much caution should be taken. In particular, it does not directly address two key questions: When is it appropriate to apply the precautionary principle? And what types of precautionary actions are warranted and at what price?

Source: Bodansky, 1991: 5.

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conducted on DDT regarding its acute toxicity, but no tests had been completed regarding its chronic effects – and it was the latter that caused problems for health. For CFCs, the properties (persistence and stability) that make it destructive to the ozone layer were judged initially to reduce the likelihood that they would cause negative environmental impacts.

While the precautionary principle is not and cannot be a panacea for complexity and uncertainty regarding environmental problems, it does provide a counterbalance to the "wait and see" attitude that often prevails. For, as Smith (1990: 112) has observed, "Scientific uncertainty is a difficult issue because uncertainty means that it is always possible to argue that better policies can be developed by waiting until a broader scientific consensus emerges".

2.5 Hedging and flexing

Hedging and flexing are strategies for decision making under conditions of uncertainty which are so severe that Collingridge (1983) characterized them as a state of "ignorance". Each is considered here.

2.5.1 Hedging

Hedging is one of the most frequently used strategies when having to make decisions under ignorance. Hedging involves a conscious choice to avoid the worst consequences by comparing all the options against the estimated worst case, and then selecting the alternative with the "least bad worst outcome". Hedging is consistent with the precautionary principle, in that it counsels caution and strives to minimize damage if the worst outcome occurs. However, Collingridge argued that while hedging is a reasonable strategy under extreme uncertainty, it is not the best strategy under conditions of ignorance. His position emphasizes the importance of being able to differentiate among conditions of risk, uncertainty and ignorance, as discussed in Chapter 1.

Collingridge concluded that hedging is not a good approach when ignorance occurs because in such situations no decision can ever be known to be the correct one. Thus, there is always the likelihood that other options than the one taken might be better, if only the planner or decision maker were aware of them.

2.5.2 Flexing

Collingridge maintained that because any decision taken in ignorance may prove to be inappropriate, the decision maker should systematically look for error, or *monitor* any decision, by continuing to scan for other options after the decision has been taken. Furthermore, because there is little point in searching for error unless there is scope for making an adjustment, he recommended that decision makers should always favour flexible options, or ones that can be revised if they are found to be inappropriate. His conclusion was that decision makers should be prepared to take flexible decisions, to monitor them and to

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modify them. The extreme caution stipulated by hedging becomes less important if there is the willingness and capacity to modify decisions in the light of new knowledge and understanding. In contrast to hedging, flexing involves seeking the best option, risking the worst outcome, but being ready to modify or reverse a decision if the worst outcome should occur.

2.5.3 Relative merits

The relative merits of hedging and flexing are:

- Hedging is an appropriate strategy if it is unlikely that a decision or an action can be reversed or significantly modified. However, if decisions can be modified or reversed, then flexing is the more appropriate approach.
- Flexing is not a prescription for inaction. Flexing provides an opportunity to achieve the best possible outcome; it is not a prescription to do nothing and hope for the best. Hedging tends to lead to preservation of the status quo, or slight variations of it, or to premature adoption of new technologies since they are viewed as the way to resolve problems.
- Flexing increases flexibility by broadening the range of options considered, unlike hedging which limits options.
- Hedging may contribute to the realization of the worst outcome because in situations of ignorance decision makers are often involved in competition with other decision makers whose intents or activities are not easy to predict. For example, if one side develops a weapon system to protect itself from what it thinks the other side might do, the development of such a system might trigger the other side to take the action which was feared as it tries to create a hedge for itself. This pattern leads to the well known phenomenon of self-fulfilling predictions.
- In many decision situations, the worst outcome may be ambiguous because a number of "worst" outcomes may be possible. In such situations it is difficult to know which option to hedge against.
- By seeking to avoid the worst outcome, hedging may also overlook real benefits that could be achieved by attaining some happier state. In contrast, flexing seeks out the best option, but recognizes that monitoring is needed so that if the worst does happen then changes can be made to limit damage.

Box 2.9 Approaches to decision making

Most principles of decision-making under uncertainty are simply common sense. We must consider a variety of plausible hypotheses about the world; consider a variety of possible strategies; favor actions that are robust to uncertainties; hedge, favor actions that are informative; probe and experiment; monitor results; update assessments and modify policy accordingly; and favor actions that are reversible.

Source: Ludwig, Hilborn and Walters, 1993: 36.

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Neither hedging nor flexing offers perfect solutions for making decisions in the face of complexity, uncertainty or ignorance. However, the review of them is intended to provide some context for the *precautionary principle*, and to encourage you to think about its strengths and weaknesses at both conceptual and operational levels. Most of the criticisms of the precautionary approach have emphasized operational issues. The above comments highlight that conceptual issues also exist, and require our careful attention.

2.6 Planning models

Different schools or models of planning have been developed. Each reflects different values, assumptions and beliefs about the nature of the world for which planning is done, and about the role of the planner. Several models are considered below, with particular regard to the way in which they address complexity and uncertainty (Box 2.10).

2.6.1 Synoptic planning

Synoptic planning, also referred to as *comprehensive rational planning*, is the dominant planning model. Most other models have been developed as a result of concern about, or criticisms of, the synoptic approach.

Synoptic planning has a number of well established steps or phases, including: (1) defining the problem, (2) establishing goals and objectives, (3) identifying alternative means to achieve the goals and objectives, (4) assessing the options against explicit criteria, (5) choosing a preferred solution and implementing it, and (6) monitoring and evaluation. These phases are linked with feedback loops, creating the possibility to incorporate changes into planning as a result of findings or experience. The assessment of options is often completed

Box 2.10 Planning under uncertainty

Where planning for the future is *feasible* (based on good data and analytical skills, continuity in the trends being extrapolated, and effective means to control outcomes), then planning is unnecessary – it is simply redundant to what already goes on. Conversely, where planning is most *needed* (where there is absence of data and skills and controls in the presence of primitive or turbulent social conditions), planning is least feasible.

Source: Hudson, 1979: 393.

. . . planning processes can be understood as addressing different conditions of uncertainty. Thus planners must assess the actual conditions of uncertainty that characterize the particular problem they are confronting and then select a style of planning that suits those conditions.

Source: Christensen, 1985: 69.

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by using methods such as benefit–cost analysis, operations research and forecasting. Quantitative analysis is often a central element of analysis.

The comprehensive rational planning model is based on the assumption that the people involved have the characteristics of *economic man* or *economic person*. This individual has the capacity to identify and rank goals, values and objectives, and can also choose consistently among them after having collected all the necessary data and having evaluated them systematically. This individual also judges alternatives against the criterion of economic efficiency, and seeks to optimize or maximize returns.

A further value associated with comprehensive rational planning is that if the planner or analyst collects enough information, completes enough analyses and studies long enough, she or he will be able to understand the situation, and therefore be able to manage or control it. However, the reality is that in many situations the necessary data are not available. Even when such data can be collected, it is not always easy to analyse them due to the existence of intangible attributes. Furthermore, the assumption that the correct problem has been identified is not always verified.

As a result of the characteristics outlined above, and highlighted in Box 2.11, the synoptic or comprehensive rational planning model does not seem well suited to deal with complexity and uncertainty. Its departure point is a belief that with enough effort and work it is possible to achieve understanding and, therefore, to be able to manage or control systems The comprehensive rational approach is unlikely to accept the concept of chaos. It also assumes capability of

Box 2.11 Problem solving

There are times, of course, when the models one has cannot accommodate the new information. One feels confused. Confusion is painful, and people strive ardently to quiet it, i.e., to make things understandable. Oftentimes, this striving manifests itself as a tenacious persistence to "get things straight." At other times, however, people jump to conclusions without adequately examining the problem. Their discomfort with the uncertainty is such that any solution will do. Moreover, if these efforts fail to bring cognitive closure, people typically respond emotionally with frustration, anger, helplessness, or apathy. Neither jumping to conclusions nor any of these affective states result in very effective problem-solving. [A study] has provided a summary of problem-solving tendencies, which, even if exaggerated, suggests the real world implications and frustrations of an inadequate problem exploration. They claim: 90% of problem solving is spent:

- Solving the wrong problem
- Stating the problem so that it can be solved
- Solving a solution
- Stating a problem too generally
- Trying to get agreement on the solution before there is agreement on the problem.

Source: Bardwell, 1991: 605.

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participants (economic person) that rarely can be met. As a result, other planning models have emerged. The most frequently considered alternative is *incremental planning*.

2.6.2 Incremental planning

The incremental planning approach has also been referred to as *disjointed incrementalism* or as *muddling through*. Instead of accepting the idea of an economic person, this approach is based on the idea that people are "boundedly rational" and who "satisfices" rather than maximize. In other words, while economic man is able to cope with the complexity of the real world, the boundedly rational person quickly simplifies the buzzing, booming confusion that characterizes the real world into a more simple model. In that sense, the world is "bounded" because not all the detail and complexity is considered. In addition, whereas the economic person strives to maximize, the boundedly rational person searches for a solution which is "good enough" or satisfactory – it does not have to be the optimum.

The incremental approach was developed to describe how things often happen in practice, but over time it also has been interpreted in a normative or prescriptive manner. That is, the model is offered as a preferred way of planning, because it captures much of the reality about the world in which planners function. Thus, in the incremental approach the planner is faced with, and recognizes, multiple problems, goals and values. She or he thus does not try to optimize, but rather to identify practically attainable goals that are generally satisfactory. All alternatives are not known, and no attempt is made to consider a broad range of options. The cost in time, effort and money to obtain additional data and to identify a broader mix of alternatives is recognized to be high.

The incremental approach thus has a number of characteristics, including:

- The problem is not clearly defined. Often, the major task for the policy maker or planner is to determine the nature of the problem to be handled.
- Goals, values and objectives may conflict with one another.
- Only a limited number of options are considered, and those differ only incrementally from each other and from the existing policy or practice.
- For each option, only a restricted number of "significant" impacts are identified.
- The problem is redefined on a regular basis. Normally, it is thought that means are adjusted to ends. Under incrementalism, the reverse often happens. Ends are modified with regard to available means.
- No single correct solution exists. Indeed, the policy maker often does not know what is wanted, but does know what should be avoided. Policies move away from bad or undesirable things, without necessarily moving systematically toward good or desirable outcomes. A satisfactory decision is one for which substantial agreement exists, even though not everyone may believe that the decision is the best relative to a given objective.
- The decision or policy process never ends. The process is viewed as a sequential chain, involving an ongoing series of incremental decisions.

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With such attributes, the incremental approach concentrates attention upon familiar and better-known experiences, limits the number of alternatives to be explored, and reduces the number and complexity of variables to be considered. The incremental planner may appear to be tentative, timid, indecisive, hesitant, cautious and narrow. At the same time, such a planner also may be viewed as realistic and pragmatic, and as a shrewd problem solver who is astute enough to recognize and accept the complexity and uncertainty surrounding environmental and resource issues. The incremental planner will deliberately choose a policy or option, knowing very well that it is not quite adequate, in order to leave open a range of other options. This path will be taken rather than selecting a policy that appears to be on target, but difficult to modify. Attention often will be given to solving smaller problems (poor seed) when it is recognized that he or she cannot solve a larger problem (low agricultural productivity). The incrementalist believes that policy making is serial or sequential, and as a result that continual nibbling is as good as taking one large bite. In this manner, the incrementalist approach is compatible with the ideas contained in the precautionary principle and in flexing. As Lindblom (1974: II–34), the architect of the incremental approach, has remarked:

> . . . most of us believe that because we became involved in our environmental difficulties piecemeal, we shall have to get out comprehensively. . . . Clearly the argument contains a fallacy. We did fall into our environmental problems through piecemeal gradualism. That still leaves open the possibility that the same route is the only route out of the problem. . . . Believing that everything is connected, we fall into the logical fallacy of believing the only way to improve those interconnections is to deal with them all at once. . . . But because everything *is* connected, it is beyond our capacity to manipulate variables comprehensively. Because everything is interconnected, the whole of the environmental problem is beyond our capacity to control in one unified policy. We have to find critical points of interventions. . . .

As with all models, the incremental approach has weaknesses. A major criticism is that the incrementalist, in believing that an evolutionary approach is best, will not consider an abrupt or radical (or revolutionary) shift in policy or practice if conditions change markedly. Since the incrementalist only considers options which are marginally different from the status quo, such a planner is unlikely to consider innovative ways significantly different from current practice. There may be situations in which a sharp change in policy or practice is needed, and the incrementalist is unlikely to be prepared to consider a radical shift. As a result, incrementalists are often characterized as being reactive to existing conditions, rather than being proactive in trying to move towards an improved state of affairs.

2.6.3 Mixed scanning

Mixed scanning was developed to capture the strengths of each of the synoptic and incremental approaches, but also to minimize their weaknesses. In particular,

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mixed scanning rejects the concept of economic person, and also rejects the aspect of incrementalism which creates an inability or unwillingness to consider fundamental changes in policy or practice. A basic idea is that many incremental decisions eventually can lead to a fundamental shift, and that the cumulative effect of many incremental decisions is influenced by fundamental decisions.

The core ideas of mixed scanning are that: (1) the decision maker relies mainly upon a continuous series of incremental decisions, but that (2) the decision maker is also steadily scanning a limited range of other alternatives, each of which represents a major departure from present practice. Thus, unlike the incrementalist, the mixed scanner looks for and considers options which are significantly different from the status quo. On the other hand, unlike the economic person who uses synoptic planning, the mixed scanner restricts attention to a limited number of these markedly different options. As a result, the mixed scanning approach is less cautious than the incremental approach, but it remains pragmatic by recognizing the cost and effort required to examine a wide variety of options.

An analogy has been made with the way in which a captain functions on a large ship, especially as it enters a new port. The attention of the captain and others on the bridge is very focused while approaching the harbour, watching for signs of shoals, small ships and other obstacles to the ship's successful entrance to the port. At the same time, the ship's radar is scanning 360 degrees, and provides information about possible storms or other inbound ships that the captain must consider in making decisions about the most suitable route to enter the port. Thus, the ship's captain is continuously scanning for distant but possibly significant aspects that need to be considered, while still keeping most attention focused on a smaller range of matters which require attention in the short-term future.

2.6.4 Transactive planning

The synoptic approach is often associated with the acceptance of expert input, and of a more centralized form of management or decision making. It usually assumes that the planners are in the best position to define the nature of the issues or problems that require attention, and to develop alternative solutions to them. In the *transactive approach*, the belief is that it is important to consider the experience of people who will be affected by the planning or decisions. Thus, planning is not a technocratic exercise conducted by experts, but rather should involve face-to-face interaction between the planners and those most affected by their activity. In the transactive approach, the key characteristics are inter-personal dialogue and mutual learning.

Transactive planning therefore transforms the role of the planner from one of distant expert to one of facilitator and participant. The planner is not assumed to hold all of the necessary or useful knowledge and wisdom, but rather is one person of many with a constructive contribution to make. Transactive planning also seeks to achieve more decentralized decision making as a means to provide more control over planning by the local people. In this manner, the process is

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consistent with the ideas of social justice, equity and empowerment advocated by sustainable development (discussed in Chapter 4).

As Hudson (1979) has commented, in contrast to incremental planning, transactive planning gives higher priority to processes of personal and organizational development rather than to the realization of particular functional objectives. As a result, plans are assessed not only in terms of what they provide regarding effective delivery of goods and services, but also to the way in which they affect local people, especially their dignity and sense of effectiveness, their values and behaviour, and their capacity for growth through cooperation. Thus, transactive planning is an approach that places a high value on creation of partnerships (Chapter 8) and on incorporating local knowledge systems into planning (Chapter 9).

2.6.5 Perspective

No single planning approach or model is perfect. Each offers strengths and weaknesses. In addition, each makes different assumptions about the role of planners and analysts, and those for whom planning or analysis is being done. It is important to recognize the merits and shortcomings of various approaches, and not to use a single approach automatically. Depending upon circumstances and conditions, one approach may be more appropriate than another. We need to have a critical appreciation of the various approaches or models, so that we can consciously choose which one is likely to be most effective in a given problem-solving situation. It also is helpful to be able to recognize when viewpoints or comments reflect acceptance of one or another model, so that it is possible to identify why people are disagreeing about a process, and to determine how to resolve such differences.

2.7 Implications

Complexity. Uncertainty. Surprise. Turbulence. All of these conditions or characteristics appear to be more the norm than the exception in resource and environmental management. It appears that resource and environmental managers would be well advised to approach their task with the expectation that surprises will be usual and change will be common. As a result, it is sensible to develop strategies and approaches that allow us to learn and adapt on the basis of experience.

The precautionary principle explicitly recognizes that our information and understanding may be inadequate, and yet that situation is not a reasonable excuse for not making decisions or taking action. In that context, when there is the possibility for reversibility of decisions, *flexing* appears to be a better approach than the more often used *hedging*. Furthermore, *mixed scanning* or *incremental* approaches to planning are more likely to be effective than a *comprehensive rational approach*. In the remaining chapters, attention will be given to various ways to deal with complexity and uncertainty, especially in situations

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characterized by rapid change and conflict. Interest also will centre on ways to move towards resource and environmental management that will be sustainable in the long term, and that will be consistent with an ecosystem approach.

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